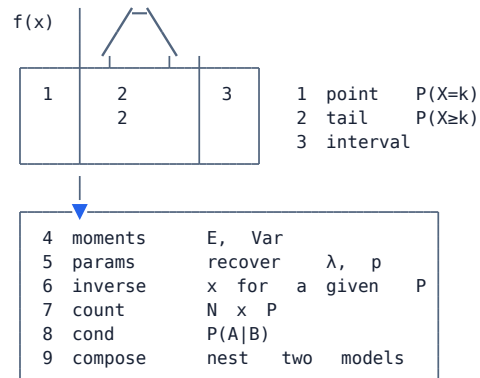


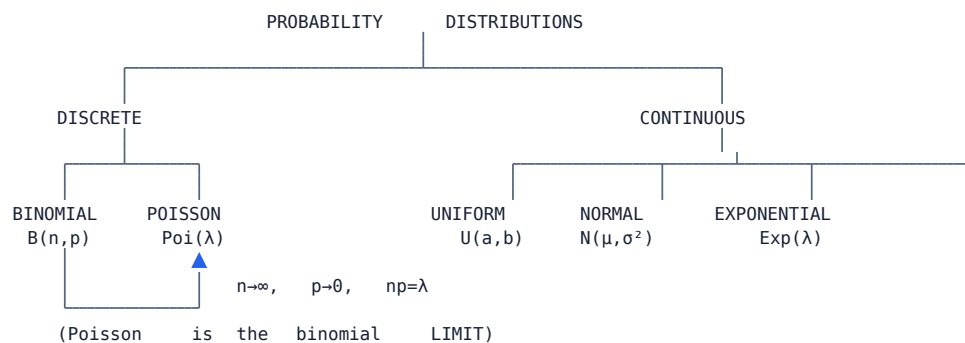
05 DISTRIBUTIONS: all five, all slots, all values

THE NINE SLOTS, VISUALISED (every distribution)

a distribution is a SHAPE, and the exam asks 9 questions of it:



THE DISTRIBUTION FAMILY TREE



MEAN vs VARIANCE, EVERY DISTRIBUTION

MEAN	VARIANCE		
$B(n,p)$	$ np$	$ npq$	
$Poi(\lambda)$	$ \lambda$	$ \lambda$	← SAME!
$U(a,b)$	$ (a+b)/2$	$ (b-a)^2/12$	
$N(\mu, \sigma^2)$	$ \mu$	$ \sigma^2$	← square
$Exp(\lambda)$	$ 1/\lambda$	$ 1/\lambda^2$	← recip
the memory hooks:			
binomial	->	np and npq	(the q is the only extra)
Poisson	->	ONE parameter does both jobs	
uniform	->	span ² / 12	
normal	->	second slot is VARIANCE, square-root it	
exponential -> reciprocal pair: rate in, time out			

For each distribution: what it models, the setup, pmf/pdf, cdf, mean, variance, sd, the standardisation if any, the table if any, and every question slot the corpus has used. The slot names are the same nine in every table so you can see the pattern (full pattern logic in 07-PATTERNS.md).

1. Binomial, $X \sim B(n, p)$

models		number of successes in n independent two-outcome trials		
setup check		two outcomes, finite n, trials independent, p constant (ppt3 p011)		
pmf		$P(X=x) = C(n,x) p^x q^{(n-x)},$	$q = 1-p,$	$x = 0..n$
cdf		$P(X \leq x) = \sum_{i=0..x} C(n,i) p^i q^{(n-i)}$		
mean		np		
variance		npq		
sd		$\sqrt{npq}$		
mode		$\text{floor}((n+1)p)$, with two adjacent modes if $(n+1)p$ is an integer		
approximation -> Poisson in the limit $n \rightarrow \infty, p \rightarrow 0, np \rightarrow \text{finite } \lambda$				
slot	ask		used in	
1 point	$P(X=2)$ out of 12, $p=0.1$		ppt3 p013-016 pens	
2 tail	$P(X \geq 2) = 1 - P_0 - P_1$		ppt3 p013-016 pens	
3 interval	$P(2 \leq X \leq 4)$		NOT asked, high risk (see 07)	
4 moments	$E=np, \text{Var}=npq$		A2 A2, A9, D3	
5 params	p from $P(X=5)=2P(X=4) \rightarrow 5/8$		ppt3 p017-018	
6 inverse	smallest n for a guarantee		NOT asked, high risk	
7 count	$N \times P$, 10000 sets $x \ (3/8)^{10} = 0.5499$		ppt3 p018	
8 conditional	$P(X=k \mid X \geq j)$		sibling drill, not an A2 B5 item	

slot	ask	used in
9 compose	Poisson per minute then binomial over 5	ppt3 p026

2. Poisson, $X \sim \text{Poi}(I)$

models	counts of rare events in a fixed interval	
setup check	rare, independent in time/space, rate constant	
pmf	$P(X=x) = e^{-I} I^x / x!$	$x = 0, 1, 2, \dots$
cdf	$P(X \leq x) = e^{-I} \sum_{i=0..x} I^i / i!$	
mean	I	
variance	I	(mean EQUALS variance, the signature)
sd	$\sqrt{I}$	
additivity	sum of independent Poissons is Poisson with I summed	
from binomial $I = np$		
slot	ask	used in
1 point	$P(X=4), I=5 \rightarrow 0.1755$	ppt3 p027 (slide prints 0.1745)
2 tail	$P(X>2) = 1 - P_0 - P_1 - P_2 = 0.8753$	A2 C3
2 tail	"rejected" = $P(X \geq 4) = 0.2424$ when $I=2.5$	A2 B1
3 interval	$P(2 \leq X \leq 4)$	NOT asked, medium risk
4 moments	$E = \text{Var} = I$, the signature test	MCQ (R25S4-A2)
5 params	I from $P(X=1)=0.2, P(X=2) \rightarrow I=10$	ppt3 p025
6 inverse	not used	low evidence
7 count	expected defective bottles 0.1 percent	E25SUM-Q17
9 compose	nest inside binomial, $32 e^{-10} = 0.00145$	ppt3 p026

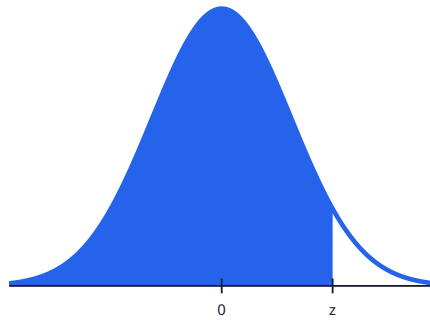
3. Uniform, $X \sim U(a, b)$

models	a random point in an interval, all lengths equally likely	
pdf	$f(x) = 1/(b-a)$,	$a \leq x \leq b$
cdf	$F(x) = 0$ below a ,	$(x-a)/(b-a)$ inside, 1 above b
mean	$(a+b)/2$	
variance	$(b-a)^2 / 12$	
sd	$(b-a)/\sqrt{12}$	
shortcut	probability = length ratio, no integration needed	
slot	ask	used in
1 point	$f(x)$ value, and mean/variance	ppt4 p003-006
2 tail	$P(X>k)$ as a length ratio	ppt4 p006
3 interval	$P(3<X<5) = 2/4 = 0.50$	ppt4 p006
4 moments	$E=(a+b)/2, \text{Var}=(b-a)^2/12$	A2 A4
5 params	solve a, b from mean and variance	A2 A10 variant
6 inverse	K such that $P(X<K)=p$	NOT asked, medium risk
9 compose	sum of two stated uniforms	notes p126 example

4. Normal, $X \sim N(\mu, \sigma^2)$

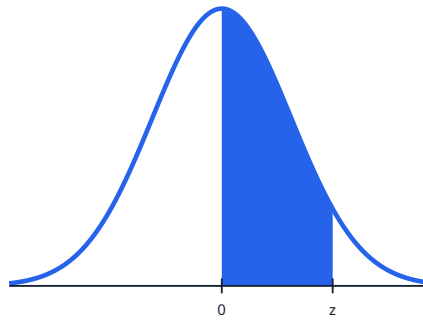
Two normal-table conventions

p022: cumulative $P(Z < z)$



$z=2: 0.9772$

p023: centre area $P(0 < Z < z)$



$z=2: 0.4772$

Two normal-table conventions

models	heights, weights, marks, measurement error, and the CLT destination	
pdf	$f(x) = 1/(\sigma \sqrt{2\pi}) \exp(-(x-\mu)^2/(2\sigma^2))$	
notation	$X \sim N(\mu, \sigma^2)$, the SECOND slot is VARIANCE	
standardise	$Z = (X-\mu)/\sigma$,	$Z \sim N(0,1)$
probability	$P(a < X < b) = \Phi((b-\mu)/\sigma) - \Phi((a-\mu)/\sigma)$	
inverse	$X = \mu + Z \sigma$	
symmetry	$\Phi(-z) = 1 - \Phi(z)$,	$P(Z > z) = 1 - \Phi(z)$
shape	mean = median = mode = μ , tails never touch, σ widens the curve	
landmarks	68.27 within 1s, 95.45 within 2s, 99.73 within 3s	
slot	ask	used in
1 point	density value, symmetry facts	ppt4 p010-011
2 tail	$P(X > 8.6) = 0.4522$	ppt4 p032-033
3 interval	$P(45 < X < 62)$, impurity=0.1645 by rounded table A2 B3, clt p013 (errata 6)	
4 moments	$E=\mu$, $\text{Var}=\sigma^2$, identify both	E25S4-A4, E25S3-A5
5 params	two unknowns from two probabilities	
6 inverse	x for a 20 percent lower tail = 3.792	ppt4 p034-037
7 count	$N \times P$, 5000 batteries, 10000 bulbs	
8 conditional	$P(X > a \mid X > \dots)$ normal ratio	not used, low evidence
9 compose	CLT $\rightarrow$ normal, then a table read	
		clt p013, p015

5. Exponential, $T \sim \text{Exp}(\lambda)$

models	waiting time between events, lifetimes, "until failure"		
pdf	$f(t) = \lambda e^{-\lambda t},$	$t > 0$	(λ is a RATE)
cdf	$F(t) = 0$ for $t \leq 0$; $1 - e^{-\lambda t}$ for $t > 0$		
tail	$P(T > t) = 1$ for $t < 0$; $e^{-\lambda t}$ for $t \geq 0$		
mean	$1/\lambda$		
variance	$1/\lambda^2$		
sd	$1/\lambda$		
memoryless	$P(T > s + t \mid T > s) = P(T > t)$		
mean-first form	$f(t) = (1/\mu) e^{-t/\mu}$ when the question gives μ , then $\lambda = 1/\mu$		
slot		ask	used in
1 point		f value, pdf identification	E24S4-A4
2 tail		$P(T > 12) = 5e^{-4} = 0.0916$	A1 short Q2 (errata 3)
3 interval		$P(T < 3\text{min}) = 0.5276$ at 15/hour	ppt4 p042-043
4 moments		$E = 1/\lambda, \text{Var} = 1/\lambda^2$	ppt4 p043
5 params		λ from a mean, or mean from λ	A2 D2
6 inverse		not used, medium risk	
7 count		not used	
8 conditional		$P(X < 1 \mid X < 2) = 0.6225$	A2 B5 (errata 9: prints 0.5679)
9 compose		repair-time, memoryless, unit conversion	A2 MCQ5, C1
ERRATUM 4: the ppt4 p040 slide says " $1/\lambda$ is the mean number per unit time", backwards.			
lambda is the rate, $1/\lambda$ is the mean TIME. p041 states it correctly.			

The one-table summary (memorise this block)

pmf / pdf	mean	variance	
$B(n,p)$	$C(n,x) p^x q^{(n-x)}$	np	npq
$Poi(l)$	$e^{-l} l^x / x!$	l	l
$U(a,b)$	$1/(b-a)$	$(a+b)/2$	$(b-a)^2/12$
$N(\mu, s^2)$	$1/(s \sqrt{2\pi}) e^{-(x-\mu)^2/2s^2}$	μ	s^2
$Exp(l)$	$l e^{-l t}$	$1/l$	$1/l^2$
memory hooks:			
binomial: np and npq , the q is the only extra			
Poisson: mean IS variance, necessary but not enough to identify the family			
uniform: centre and span squared over 12			
normal: μ and σ^2 , the second slot is variance			
exponential: reciprocal pair, rate in, time out			